

POPESCU Vlad-Daniel

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EDUCATION

POLITEHNICA Bucharest

Sept 2024 — Present

B.Sc. in Computer Science (ACS CTI)

RELEVANT COURSEWORK

Algorithm Analysis, Data Structures, Java, Scala, Functional Programming, Computer Networks, Communication Protocols, Operating Systems, Assembly Language & Computer Architecture

Tudor Vladimirescu
National College

2020 — 2024

Baccalaureate: **10 / 10** in Physics; outstanding results across all subjects.

SKILLS

Languages: C, C++, Bash, Java, Scala, Python, JavaScript, Haskell, x86 Assembly, MATLAB

Build & Debug Tools: GCC, Makefile, Valgrind, GDB, Maven

Developer Tools: Git, GitHub, VS Code, Linux, Remote Access Tools

Concepts: Memory Management, Systems Programming, Unikernel Architecture, Compiler Design, Data Structures & Algorithms

Scripting & Automation: Bash scripting, CLI tool development

EXPERIENCE

Actively participated in numerous workshops and hackathons, gaining hands-on experience in competitive and collaborative environments.

Google Student AI Hackathon

2026 · Bucharest

3rd Place — Competed among top student teams in a Google-organised AI challenge, delivering an innovative solution under time constraints.

PERSONAL PROJECTS

TicketHub

Java, Git, GitHub, IntelliJ IDEA, Linux

- Software Architecture:** Implemented the Builder, Strategy, Factory, and State patterns to manage complex ticket lifecycles and enable intelligent resource allocation.
- Resource Management:** Developed a filtering engine to match developers to tasks based on expertise, seniority, and availability.
- Access Control:** Integrated a role-based permission system (Manager, Developer, Reporter) with secure data persistence in JSON format.
- Code Quality:** 100% compliance with Clean Code standards and rigorous input validation.

PoliTCP

C++, C, Make, Git, GitHub, Linux

- Protocol Design:** Custom reliable transport protocol over raw UDP sockets, replicating core TCP mechanisms (sliding window, flow control, ordered delivery) at the application layer via the POSIX API.
- Flow Control:** Bidirectional sliding window on both sender and receiver sides, with BDP-aware window sizing to maximise link utilisation.
- Reliability:** Guaranteed in-order, duplicate-free delivery through cumulative ACKs, retransmission logic, and out-of-order segment buffering.
- Concurrency:** Background POSIX threads handle ACK processing and retransmissions independently, protected by per-connection mutexes.

Chaos Court

JavaScript, CSS, Node.js, Git, GitHub

- Origin:** Built at the 2026 Google Student AI Hackathon (Bucharest) — **3rd place** — with further features added post-competition.
- Concept:** Multiplayer debate game where players argue for or against an absurd statement before an AI judge that adopts randomised personas (1800s wizard, astronaut, etc.) and delivers a ruling.
- AI Integration:** LLM evaluates arguments in real time and generates an image based on the verdict; supports text and voice input.
- Stack:** Full-stack JavaScript with a Node.js backend, deployed on Google Cloud.